Shaheed Zulfikar Ali Bhutto Institute of Science & Technology

COMPUTER SCIENCE DEPARTMENT

Total Marks: 4

Obtained Marks: ________

Introduction to Data Science

Assignment # 04

Project Title:

AQI-Based Adaptive Traffic-Control System Using Reinforcement Learning

GitHub Repository Link: https://github.com/Abdulrehmanshabir/AQI-Data-Science-project

Last date of Submission: 8th June, 2026

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AQI-Based Adaptive Traffic-Control System Using Reinforcement Learning

Part A: Continuation from Assignment 3

This assignment is a continuation of Assignment 3. The cleaned AQI dataset, AQI values, AQI categories, and exploratory data analysis results from Assignment 3 were reused. The same GitHub repository and dataset were used as the environment for the reinforcement learning model.

Part B: Reinforcement Learning Components

RL Component	Definition
Agent	Adaptive Traffic-Control Decision Maker
Environment	Cleaned AQI Dataset
State	Low AQI, Medium AQI, High AQI
Action	No Restriction, Partial Restriction, High Pollution Alert
Reward	Positive reward for correct action and penalty for incorrect action
Policy	Best action selected for each AQI state
Episode	One complete training cycle

Part C: AQI State Creation

To simplify the reinforcement learning problem, AQI values were converted into three states. AQI values from 0–100 were labeled as Low AQI, values from 101–200 were labeled as Medium AQI, and values above 200 were labeled as High AQI.

AQI Range	RL State
0 – 100	Low AQI
101 – 200	Medium AQI
201+	High AQI

Part D: Action Space

The reinforcement learning agent was given three possible actions. These actions represent different levels of traffic control based on air quality conditions.

Action ID	Action Name	Description
0	No Restriction	Normal traffic flow without any restrictions
1	Partial Restriction	Moderate control measures for pollution reduction
2	High Pollution Alert	Strong warning and strict traffic restrictions

Part E: Reward System

A reward system was designed to encourage the agent to select suitable traffic-control actions for different AQI conditions. Positive rewards were assigned to correct actions, while penalties were assigned to inappropriate actions. This helps the agent learn an effective policy that balances public health protection and traffic flow.

AQI State	No Restriction	Partial Restriction	High Pollution Alert	Best Action
Low AQI	+10	-2	-5	No Restriction
Medium AQI	-6	+10	+2	Partial Restriction
High AQI	-10	-3	+10	High Pollution Alert

Part F: Q-Learning Implementation

A Q-learning algorithm was implemented to train the agent. The agent learns optimal actions using reward feedback.

Key parameters used:

- Learning rate (alpha): 0.1
- Discount factor (gamma): 0.9
- Exploration rate (epsilon): 0.2
- Episodes: 500

The Q-learning update rule updates the Q-table based on experience:

$$Q(s,a) = Q(s,a) + \alpha \, [\, reward + \gamma \max(Q(s',a)) - Q(s,a) \,]$$

After training, the Q-table represents the learned policy.

Part G: Simulation and Evaluation

After training, the model was evaluated using the learned policy.

Results:

- The agent learned optimal actions for each AQI state.
- Average reward improved over episodes.
- The Q-table showed convergence toward stable values.

Learned Policy:

- Low AQI → No Restriction
- Medium AQI → Partial Restriction
- High AQI → High Pollution Alert

The learned policy closely matches logical real-world decisions.

Part H: Required Visualizations

Three visualizations were used to evaluate the RL model:

1. Reward per Episode

This graph shows how the total reward changes over training episodes.
An increasing trend indicates that the agent is learning better decisions over time.

2. Q-Table Heatmap

This visualization shows the learned value of each action for every state.
Higher values indicate better actions for that state.

3. Action Distribution

This chart shows how frequently each action was selected during training.
It helps to understand the behavior of the trained agent.

5. Final RL Result Tables

Metric	Value
Number of States	3
Number of Actions	3
Episodes	500
Learning Rate (α)	0.1
Discount Factor (γ)	0.9
Exploration Rate (ε)	0.2
Average Reward	(from notebook output)
Final Policy	Low=0, Medium=1, High=2

8. Questions Students Must Answer

1. What is reinforcement learning?

Reinforcement learning is a machine learning method where an agent learns by interacting with an environment and receiving rewards or penalties.

2. What is the agent in this AQI traffic-control problem

The adaptive traffic-control system.

3. What is the environment in this problem?

The cleaned AQI dataset from Assignment 3.

4. Which AQI states did you define and why?

Low, Medium, and High AQI based on pollution levels.

5. What are the three actions available to the agent?

- No Restriction
- Partial Restriction
- High Pollution Alert

6. How did you design the reward system?

Positive reward for correct actions and negative reward for incorrect decisions.

7. What does exploration vs. exploitation mean?

- Exploration = trying new actions
- Exploitation = using best known action

8. What did the final Q-table show?

It showed learned values of each action for each AQI state.

9. Which action did the agent learn for Low, Medium, and High AQI?

- Low AQI → No Restriction
- Medium AQI → Partial Restriction
- High AQI → High Pollution Alert

10. Did the RL agent make logical decisions? Explain.

Yes, the agent learned environmentally correct traffic control policies.

10. Did the RL agent make logical decisions? Explain.

- Simplified states
- No real traffic simulation
- Small dataset assumptions
- Fixed reward system